

# Metro Re-illustrated: Incremental Generation of Stylized Paintings Using Neural Networks

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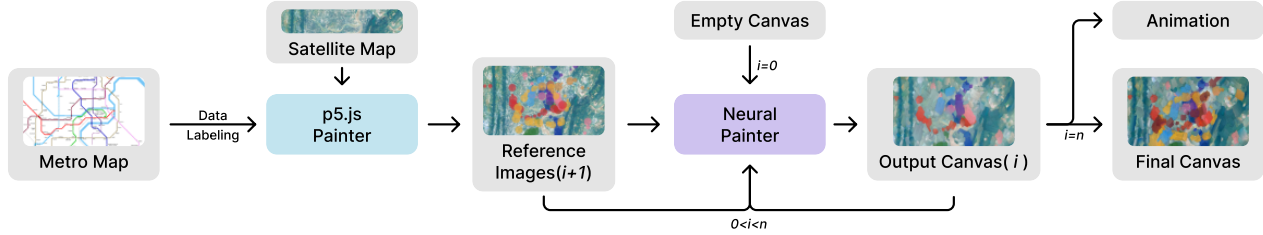


Figure 1: The workflow of Metro Re-illustrated

## ABSTRACT

Metro Re-illustrated is a project that explores incremental generation of stylized paintings of city metro maps using neural networks. It begins with an interactive system for labeling time-series data on city metro maps and generating reference images. These images are fed into a neural painter that incrementally generates oil painting-like strokes on virtual canvases. The generated paintings demonstrate blending and layering features of oil paintings while capturing the progressive nature of urban development.

## CCS CONCEPTS

• Applied computing → Media arts; • Human-centered computing → Visualization systems and tools.

## KEYWORDS

stylized painting, metro map, stroke generation,

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## 1 INTRODUCTION

Stylized painting generation has become a popular area of research in computer vision, as it combines artistic expression with advanced

algorithms to create visually striking images. This project explores incremental generation of stylized paintings for city metro maps using neural networks. First, An interactive system is developed to label time-series data of city metro railway systems and generate reference images for oil paintings generation. Then, The labeled data are structured into a novel dataset with uniform format for city metro, facilitating the project and providing resource for future research. Finally, we implement incremental generation of strokes on a virtual canvas, exhibiting blending and layering features of oil paintings as well as the city development process in animation.

## 2 THE METRO RE-ILLUSTRATED SYSTEM

### 2.1 Workflow

The workflow of Metro Re-illustrated is shown in Figure 1. We utilize a p5.js painter to label metro data on the transit maps and incorporate them onto the satellite maps to create detailed reference images that serve as input for the neural painter.

The neural painter takes these reference images as input and generates oil painting strokes on an empty canvas. Resulting painting of each reference is used as the base canvas for the next iteration. This iterative process continues until all reference images have been painted, creating a series of incrementally generated paintings, which are later encoded into an animation video.

### 2.2 Data Processing and Reference Image Generation

We developed a data processing program using a toolkit for creating graphics and interactions called p5.js[Foundation 2023], offering map retrieval and data labeling functionality in the browser, as depicted in Figure 2(a). The system features a labeling map on the left and control buttons with a status display area on the right.

In the labeling process, when clicking on metro stations on the map, the coordinate information are temporarily stored in the program. Users can navigate forward or backward through lines until

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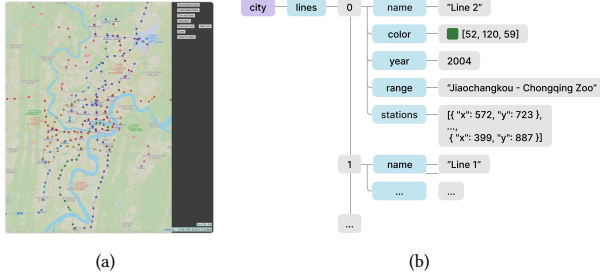
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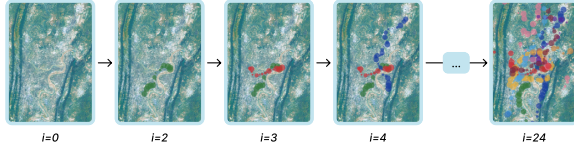
all lines are labeled. Upon completion, users can visualize the annotated data on the map and export a JSON file of all the data.



**Figure 2: Labeling tool and structure of city metro data**

The data is organized by city, with the data structure displayed in Figure 2(b). Each line contains information such as the line's name, representative color, start year of operation, operating interval, and the coordinates for each station along the route. It is worth noting that many metro lines are constructed and operated in segments synchronizing with the pace of urban development [Lin et al. 2022]. Consequently, the dataset may contain multiple records with the same name but different years.

Labeled data are then merged with the satellite map to create reference images for the painting generation, as illustrated in Figure 3. We draw the stations at their corresponding coordinates on the map as randomly sized circles of the corresponding line color. In the drawing process, we employ a time-based strategy. The current image is saved after drawing each line, ensuring a clear hierarchy and distinction of lines and changes within the reference images.

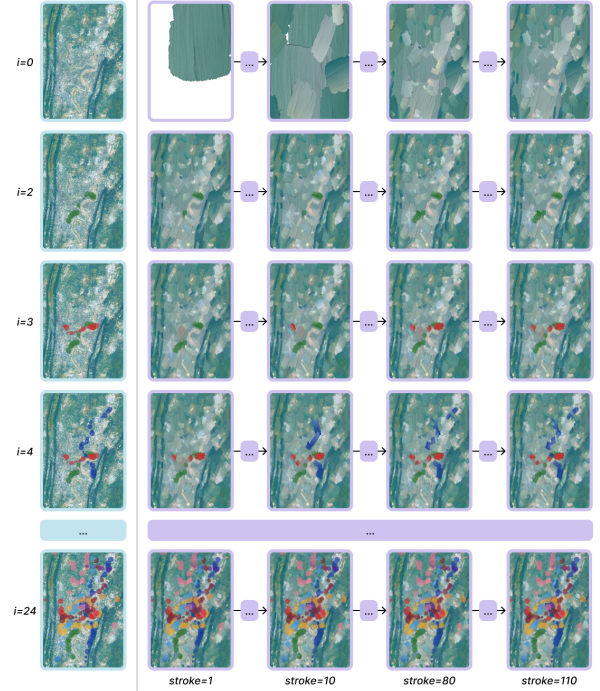


**Figure 3: Reference images generation process, where  $i$  is the number of metro lines.**

### 2.3 Stroke Generation and Animation Encoding

The stroke generation is completed by a neural painter which applies strokes on an virtual canvas and calculates the loss between the generated image and the reference image to update the parameters [Zou et al. 2021]. In the stroke generation process shown in Figure 4, we start with a blank canvas and first try to paint the satellite map image, then in the next iterations, the blank canvas is replaced with the generated painting from the previous iteration and try to paint the next reference image. This process is repeated until all the reference images have been painted.

For each reference image, we iteratively generate 110 strokes on the canvas. The canvases after stroke 1, 10, 80 and 110 are shown in Figure 4. Since we use the generated image from the previous iteration as the canvas for the next, the painter utilizes the background layer to predict strokes. The sequence of frames are then encoded into an animation to show the painting process. The resulting strokes exhibit color blending and stroke layering features between the background and new reference images, which enhances the authenticity of the generated oil painting.



**Figure 4: Strokes generation process, where  $i$  is the number of metro lines.**

## 3 SUMMARY AND FUTURE WORK

In summary, we explored generating incremental stylized paintings generation of metro maps in this project. Combining the power of neural networks with the artistry of stylized painting generation, our work has the potential to inform future work in data visualization and artistic applications of machine learning algorithms.

Currently, the reference images are randomly generated, which limits the information conveyed. By incorporating additional data such as passenger flow and the number of routes, the trade-off between aesthetics and readability of the visualization could be explored through user studies. Additionally, the stroke generation process could be refined by constraining the behaviour of the neural painter. This would lead to higher quality and more accurate stylized paintings. Lastly, more available public data could result in less labeling interaction steps and more dimensions of information in the visualization, enabling users to explore the city metro network more effectively.

## ACKNOWLEDGMENTS

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